



復旦大學
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It from Anomalies

Universal Defect Energy and Entropy from Weyl Anomalies

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Based on [**ZXH**-Yuan-Zhou, **Phys. Rev. Lett.** 136 (2026) 201601.]

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Motivation

- **Weyl anomaly** (central charge) governs the **ground state energy** and **entanglement structure** of CFTs

$$\text{2D CFT} \quad \langle T^\mu_\mu \rangle = -\frac{c}{24\pi} R \quad E = -\frac{c}{12} \quad S_n = \frac{c}{6} \left(\frac{1}{n} + 1 \right) \log \frac{l}{\epsilon}$$

- **Defects**: usually defined by **non-local operators/B.C.** in QFT
 - **Nontrivial interaction** with the bulk D.o.F.
 - **Surface defect Weyl anomaly**

$$\langle t^\mu_\mu \rangle = -\frac{1}{24\pi} \left[b R^\Sigma + d_1 \Pi^\mu_{ab} \Pi^\mu_{ab} - d_2 W_{ab}^{ab} \right]$$



With defect,
analogous relation?

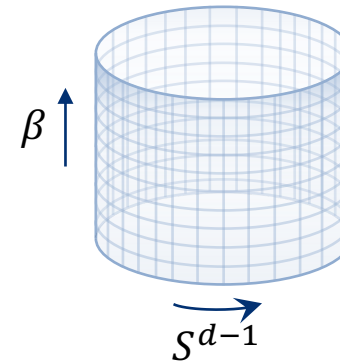
Main results

Defect Weyl anomaly governs defect contribution to the SUSY Casimir energy (dSCE) and Renyi entropy (dSRE)

➤ $\frac{1}{2}$ -BPS Wilson surface in 6d (2,0) SCFT [**ZXH**-Yuan-Zhou, 2026]

$$E_{\mathcal{R}} = -\frac{1}{\omega_1} \left[\frac{d_2 - b}{6} \omega_2 \omega_3 + \frac{2b - d_2}{24} \sigma_1 \sigma_2 \right]$$

$$S_n = \frac{2b - d_2}{12} \left[r_1 r_2 \left(\frac{1}{n} - 1 \right) + 2 \right] \log \frac{l}{\epsilon}$$

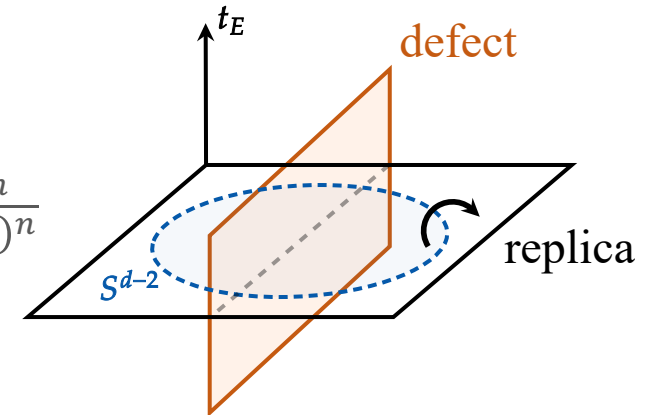


Low-temperature limit of p.f.

$$Z = \text{Tr} \left[e^{-\beta \sum_{i=1}^3 \omega_i J_i + \sum_{a=1}^2 \sigma_a R_a} \right]$$

$$\log Z = -\beta E + O(\beta^0), \quad \beta \rightarrow \infty$$

$$S_n = \frac{1}{1-n} \log \frac{Z_n}{(Z_1)^n}$$



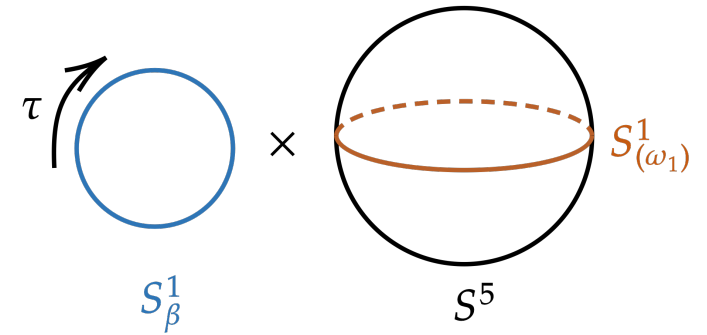
Non-perturbative method

➤ Localization

Partition function with defect: Wilson loop in 5d maximal SYM

$$\log \langle D \rangle = -\beta E_{\mathcal{R}} + O(\beta^0), \quad \beta \rightarrow \infty$$

$$\langle D \rangle = \frac{1}{Z} \int dv \exp \left[\frac{2\pi}{\omega_1 \omega_2 \omega_3} \left(-\frac{\pi}{\beta} \sum_{i=1}^N v_i^2 + \frac{\sigma_1 \sigma_2}{2} \sum_{j<i} (v_i - v_j) \right) \right] \text{Tr}_{\mathcal{R}} \exp \left[\frac{2\pi v}{\omega_1} \right]$$



➤ Holography

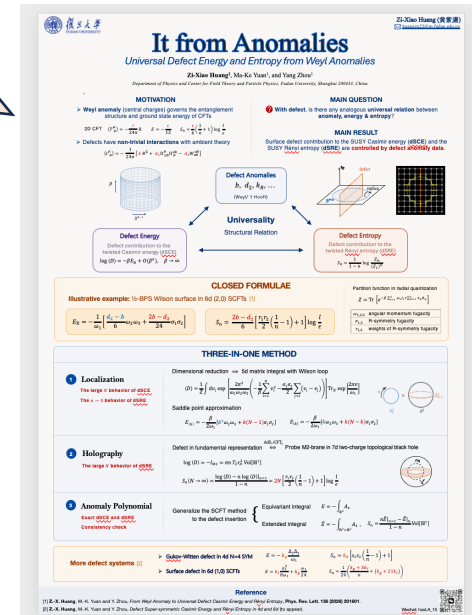
Probe M2-brane in 7d two-charge topological black hole

$$\log \langle D \rangle = -I_{M2} \quad S_n(N \rightarrow \infty) = \frac{\log \langle D \rangle - n \log \langle D \rangle|_{n=1}}{1-n}$$

➤ Anomaly polynomial

Energy/Entropy from the equivariant integral $\int A_4$

Welcome to
my poster
for details!



More defect systems [WIP]

- Gukov-Witten defects in 4d $\mathcal{N} = 4$ SYM

$$E = -k_R \frac{\Delta_2 \Delta_3}{\omega_1} \qquad S_n = k_R \left[s_1 s_2 \left(\frac{1}{n} - 1 \right) + 1 \right]$$

- Surface defect in 6d (1,0) SCFTs

$$E = k_I \frac{a_I^2}{8\omega_1} + k_g \frac{\omega_1}{24} \qquad S_n = \frac{1}{24} \left(\frac{k_g + 3k_I}{n} + (k_g + 21k_I) \right)$$

- $p = 4$ defect in 6d SCFTs, ...

Conclusion

- **Generalized the relation of anomaly, energy & entropy to defect system**

Defect contribution to SCE and SRE are controlled by defect Weyl anomaly

$$E_{\mathcal{R}} = -\frac{1}{\omega_1} \left[\frac{d_2 - b}{6} \omega_2 \omega_3 + \frac{2b - d_2}{24} \sigma_1 \sigma_2 \right]$$

$$S_n = \frac{2b - d_2}{12} \left[r_1 r_2 \left(\frac{1}{n} - 1 \right) + 2 \right] \log \frac{l}{\epsilon}$$

- **Non-perturbative** methods combining localization, holography & anomaly polynomial